

HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT

ATTACHMENT D

**PAGES FROM POWERS-RAWL FASTENING SYSTEMS DESIGN MANUAL,
SECOND EDITION**

p. 67

p. 145

p. 146-149

p. 159

p. 164

(9 Pages)

Dimensions For UNC Coarse Thread Series

Nominal Thread Size (inches)	Threads Per Inch	Basic Major Diameter (inches)	Gross Nominal Area (sq. ins.)	Tensile Stress Area (sq. ins.)
No. 6	32	0.1380	0.0149	0.0091
No. 8	32	0.1640	0.0211	0.0140
No. 10	24	0.1900	0.0283	0.0175
No. 12	24	0.2160	0.0366	0.0242
1/4	20	0.2500	0.0491	0.0318
5/16	18	0.3125	0.0767	0.0524
3/8	16	0.3750	0.1104	0.0775
7/16	14	0.4375	0.1503	0.1063
1/2	13	0.5000	0.1963	0.1419
9/16	12	0.5625	0.2480	0.1820
5/8	11	0.6250	0.3070	0.2260
3/4	10	0.7500	0.4420	0.3340
7/8	9	0.8750	0.6010	0.4620
1	8	1.0000	0.7850	0.6060
1-1/8	7	1.1250	0.9940	0.7630
1-1/4	7	1.2500	1.2270	0.9690
1-3/8	6	1.3750	1.4840	1.1550
1-1/2	6	1.5000	1.7660	1.4050
1-3/4	5	1.7500	2.4040	1.8990
2	4-1/2	2.0000	3.1400	2.4980

22.5.1 Allowable Steel Strength

In some cases, it may be desirable to calculate the allowable steel strength for a bolt or a threaded anchor rod. One method to calculate the allowable steel strength is based on the stresses as listed in the American Institute Of Steel Construction (AISC) Manual Of Steel Construction, Allowable Stress Design. Using this Method, the allowable tensile stress, F_t , and the allowable shear stress, F_v , are calculated as follows:

$$F_t = 0.33 \times F_u$$

$$F_v = 0.17 \times F_u$$

Where F_u = minimum specified ultimate tensile strength for the steel material.

This stress is then applied to the gross (nominal) area of the threaded section to calculate the load in pounds.

In addition to the load capability of the material, an anchor should be manufactured from material which is compatible with its intended use. For example, anchors manufactured from a material with a melting point of less than 1000° F. are not normally recommended for overhead applications due to fire code regulations unless specific fire rating tests have been performed. Special materials may be required for corrosive environments and galvanic reactions.

Powers Rawl reserves the right to use alternate anchor materials which will perform in a similar manner depending upon production requirements.

22.5.2 Material Certifications

Powers Rawl provides the following types of certification for products when requested by the user.

22.5.2.1 Certificate Of Compliance

This type of certification, sometimes called a Certificate Of Conformance, lists the materials and plating used in the manufacture of a product referencing pertinent specifications such as AISI, ASTM, UL, FM, or ICBO. All major components are described including nuts and washers. This is the most commonly requested type of certification. A Certificate Of Compliance can be requested for any Powers Rawl product from the local Branch office.

22.5.2.2 Mill Certifications

Requests for Mill Certifications usually apply to steel anchors. Mill Certifications provide full traceability of a finished product back to the original lot of steel from which it was produced and usually include the heat number, material identification, chemical analysis, and physical properties.

In order to produce a part which is traceable back to the original Mill Certification, the raw steel material must be identified at the start of the manufacturing process. Powers Rawl is able to perform this type of service, however, these certifications can be supplied only on material that is ordered as a special. A price and delivery quotation for any item requiring Mill Certifications can be obtained by contacting the local Powers Rawl Branch office.

22.6 Design Recommendations

22.6.1 Allowable Load Capacities

The allowable load which may be applied to an anchor is calculated based upon applying a safety factor to the average ultimate load capacity obtained from testing. One purpose of a safety factor is to allow for field variations which may differ from the testing conditions in the laboratory. Examples of these variations include differences in the type and strength of base material, the setting method used, and long term performance factors. The standard established by the industry is to reduce the ultimate load capacity by a minimum safety factor of 4 to calculate the allowable working load. For example, an anchor which has an average ultimate tension load capacity of 12,000 pounds would have a maximum allowable working load anchor of 3,000 pounds. Critical applications such as overhead applications or dynamic loading may require safety factors of 10 or higher. The allowable loads are recommendations, however, local building codes should be consulted to determine the required safety factors.

51.6 Adhesive Anchor Installation Guidelines

As with any building component, proper installation is the key to a successful application once the anchor has been properly selected. While each individual adhesive product section provides specific installation instructions, the following summary highlights general areas of importance.

51.6.1 Anchor Holes

A properly drilled hole is a critical factor both for ease of installation and optimum anchor performance. The anchors selected and the drill bits to be used should be specified as part of the total anchoring system. Powers *Rawl* adhesive anchors are designed to be installed in holes drilled with carbide tipped bits meeting the requirements of the American National Standards Institute (ANSI) Standard B212.15 unless otherwise specified. If alternate bit types such as diamond tipped core bits are used, the tip tolerance should be within the ANSI range. A diamond tipped bit drills a hole which has very smooth walls. Although adhesive anchors tend to work acceptably in diamond cored holes, testing should be conducted to verify performance.

When using adhesives, anchor holes should be thoroughly cleaned prior to installation of the adhesive. During the drilling process, dust is often pressed into the walls of the hole. Blowing the hole with compressed air or vacuuming alone will not properly clean the hole. In order to insure that the proper bond is developed with the base material, the holes should also be brushed using a nylon brush to remove dust and other debris which may have been pressed into the walls of the hole.

51.6.2 Use Of Adhesives In Cold Weather

One of the differences between epoxy and ester based resins is the type of chemical reaction. When the individual components of an epoxy are combined or mixed, the reaction which occurs is called an addition reaction. The reaction is described in this manner because it involves the addition of an epoxy molecule with an amine molecule which are then cross linked three dimensionally to form a polymer. When the temperature is warmer, the molecules are freer and able to cross link faster. As the temperature lowers, the cross linking becomes more difficult and will stop at a certain point. In house test have shown that the fast set version of the Power-Fast epoxy can gel and set in materials as cold as 25° F., however, special installation procedures should be followed.

Ester based resins react differently. In an ester based resin, the hardening catalyst causes the ester molecules or monomers to link together forming long chain polymers. In this type of reaction, the mix ratio is not as critical. While colder temperatures do not effect the reaction of these resins as severely, proper conditioning is still required.

As the temperature of the unmixed components of an adhesive material decreases, they thicken which can make dispensing or installation difficult in cold weather. To prevent difficulty during installation, Powers *Rawl* recommends that adhesives be conditioned to a minimum temperature of 60°F. prior to installation unless otherwise recommended in the individual product sections. When an adhesive is conditioned, it should be maintained at the required minimum temperature for a sufficient amount of time to insure that the entire cross section of the container is brought to temperature. Some systems such as Power-Fast Epoxy Injection Gel have tool heaters available to keep the adhesive warm.

Some products, especially epoxies, are not recommended for use in base materials having a temperature of 40° F. or less unless job site performance tests are conducted. Although these materials may gel and cure at lower temperatures, 40° F. was selected as a safe minimum temperature because experience has shown that no special installation procedures are required beyond our published instructions.

When adhesive anchors are installed in concrete which is in the freezing range, frost or ice can form on the walls of the anchor hole. If this occurs, injection type adhesives may not properly bond to the walls of the anchor hole. Spin-in type capsule systems which scrape the walls of the anchor hole during installation are less sensitive to this. Since concrete is porous and acts like a sponge, even the pour water can freeze and prevent the adhesive from properly wetting the surface of the hole. This can be prevented by heating the anchor hole with a heat gun prior to installation of the adhesive. A torch should never be used because it carbonates the concrete on the walls of the anchor hole creating a residual dust.

Since variations in the concrete mix design may effect the formation of frost or ice on the walls of an anchor hole, job site performance tests are recommended for installations of epoxies in base materials having a temperature of less than 40° F. to verify gel and cure times.

52.0 Power-Fast™

52.1 Introduction

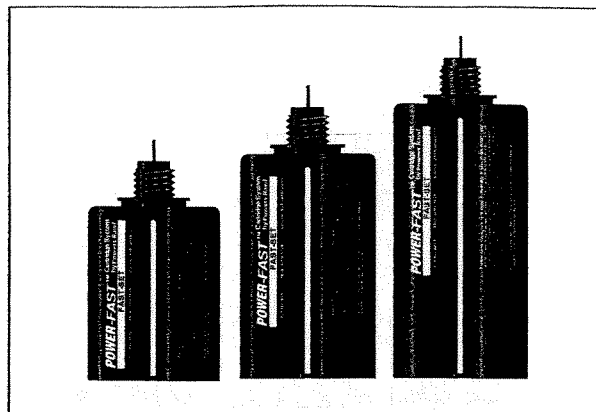
The **Power-Fast** Epoxy Injection Gel is a two component structural epoxy which is available in two dispensing systems. For small to medium size jobs, it is packaged in specially designed cartridges made from an engineered plastic which are used with either a manual or pneumatically operated injection tool. It is also packaged in reinforced foil pack sausages which are dispensed from specially designed high pressure manual or pneumatic injection tools. The epoxy used is an odorless, premium, high strength epoxy which has a proven track record. **Power-Fast** Epoxy Injection Gel is designed for use in anchoring threaded rod, bolts, reinforcing bars, and smooth dowels into concrete and other solid base materials. The system can also be used to anchor into hollow masonry materials using threaded rod and screen tubes. In addition to anchoring applications, **Power-Fast** Epoxy Injection Gel can be used for bonding steel and cured concrete to cured concrete. It can also be used for pick proofing applications, for surface sealing cracks, and for placing injection ports. Details of all system components including the **Power-Fast** Epoxy Injection Gel, dispensing tools, installation instructions, ultimate load capacities, and allowable load capacities are described in this section.

52.2 Product Description

52.2.1 Power-Fast™ Cartridge

Power-Fast Epoxy Injection Gel is offered in two dispensing systems to meet the installation requirements for most applications. The cartridge system is suitable for small to medium jobs while the foil pack version is preferred for larger jobs where faster dispensing of material in larger volumes is required. **Power-Fast** Epoxy Injection Gel is available in either fast and slow setting versions to provide the installer with a choice of gel and curing times. For example, in warmer weather, the slower setting version is normally used as it allows the installer time to properly fill anchor holes prior to the point at which the material begins to gel or solidify.

The design of the **Power-Fast** cartridge eliminates the leakage and dispensing problems often found on other products. Supplied in 10, 15 and 24 fluid ounce capacities, each cartridge is manufactured in a side by side configuration using two parallel tubes. One tube contains the Component A - Base Resin while the other tube contains the Component B - Hardener.

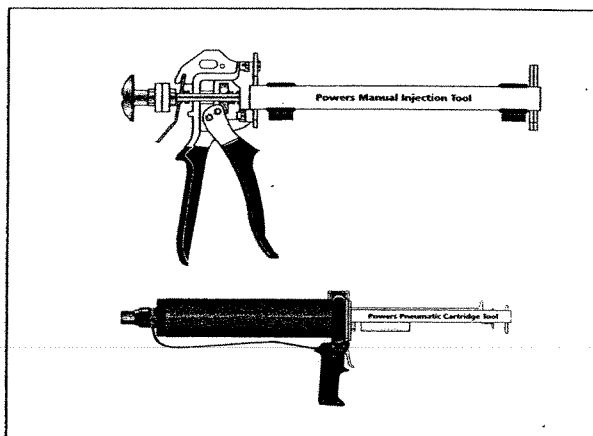


The tubes are separated by an air space to keep the components out of contact until they are ready to be dispensed.

Each cartridge is formed from an engineered plastic using a special high tolerance molding process. At the front end of each cartridge is a reinforced retention guide and a threaded manifold. To prevent leakage of the cartridge during storage, a sealing plug is inserted in the ports of the manifold, followed by a molded cap which is threaded on. A set of plastic pushers is inserted into the rear of each tube to pressurize the epoxy during dispensing. The pushers are reinforced and contain an "O" ring to prevent leakage during storage and dispensing. This provides the user with a cartridge which is easy to dispense, yet durable.

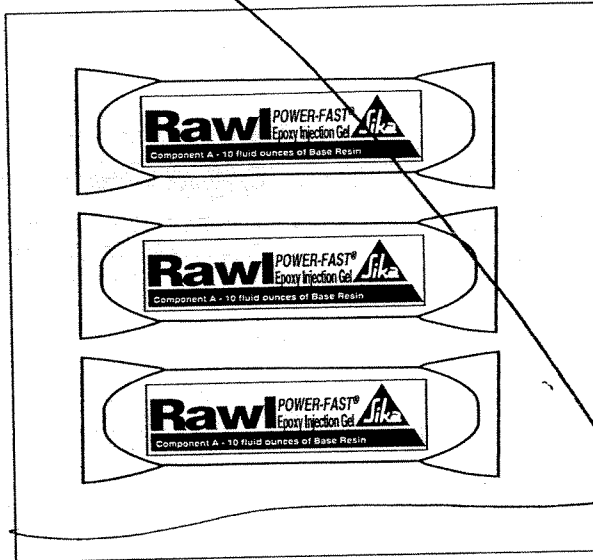
52.2.2 Power-Fast Cartridge System Dispensing Tools

Both sizes of **Power-Fast** cartridge can be dispensed from manually or pneumatically operated injection tools. Each tool has a dual piston design which applies even, consistent pressure to the cartridge to insure proper mixing, eliminating the possibility of measuring errors.



52.2.3 Power-Fast Sausage

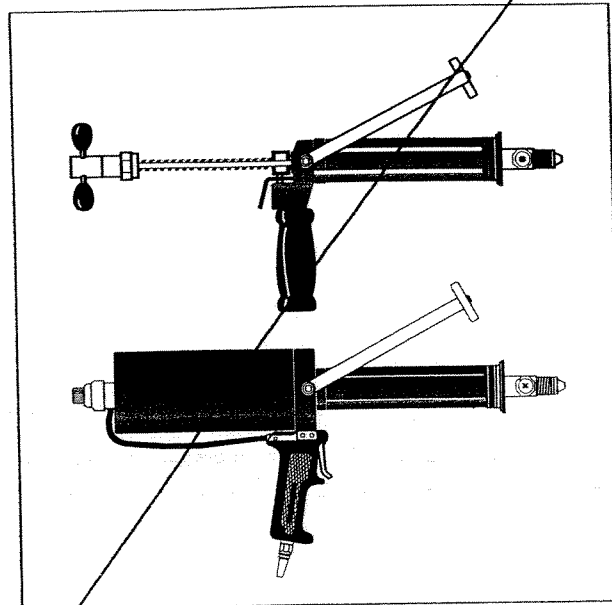
In the sausage system, the **Power-Fast™** Epoxy Injection Gel is packaged in individual mylar reinforced foil sausages which contain 10 fluid ounces of Component A - Base Resin and Component B - Hardener. The components are mixed in a 1 to 1 ratio using either a manual or pneumatic tool system. The total amount of mixed epoxy available for dispensing is 20 fluid ounces.



Each foil pack is labeled and color coded for easy identification of the components. The Component A - Base Resin foil pack is white, the Component B - Fast Set Hardener foil pack is red, and the Component B - Slow Set Hardener foil pack is blue.

52.2.4 Power-Fast Sausage System Dispensing Tools

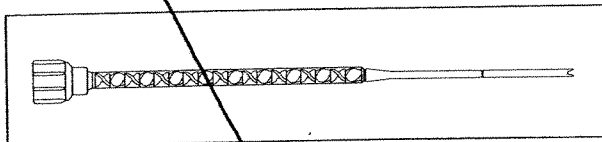
The **Power-Fast** sausages can be dispensed from specially designed manual or pneumatic injection tools. Each tool is equipped with aluminum guide cylinders which house the foil sausages. Simply insert the Component A - Base Resin into the cylinder marked A and either the Component B - Fast Set Hardener or the Component B - Slow Set Hardener into the cylinder marked B. The tools are supplied with a reusable manifold assembly which automatically opens the foil sausages and delivers the components to the mixing nozzle. With the foil packs housed in dual cylinders, maximum dispensing pressure can be applied for fast injection of the epoxy into anchoring holes because the aluminum cylinders can withstand a greater pressure than the plastic cartridges. This makes the sausage system the ideal choice for larger jobs. The dual cylinder design applies even, consistent pressure to the sausages to insure proper mixing, eliminating the possibility of measuring errors.



52.2.5 Power-Fast Mixing Nozzles

To insure complete and proper mixing of the epoxy components, the **Power-Fast** Systems use a static mixing nozzle. This reduces the possibility of mixing errors which are common with hand mixed materials. The nozzles have been designed with a unique tapered extension tip to allow one nozzle to be used to inject epoxy into most anchor hole sizes. Each nozzle contains a series of stationary components called mixing elements. The elements are motionless and remain in a fixed or static position as the epoxy components are pumped through the nozzle.

As the components are pumped through the nozzle, they are progressively divided and recombined by the stationary mixing elements to insure precise automatic mixing of the components. To ensure proper mixing, **Power-Fast** nozzles should be used.



Detailed information on the injection tools, mixing nozzles, and accessories is listed in the **Power-Fast** Cartridge and Sausage System selection guides.

52.3 Power-Fast Epoxy Material Properties

The **Power-Fast** epoxy is an amine based material which meets the requirements of ASTM C-881 Types I, II, IV, and V, Grade 3, Classes B and C. The benefits of using an amine based material include no foul odor, better resistance to moisture, and better chemical resistance. The properties listed in this section apply to the **Power-Fast** Epoxy Injection Gel based on the American Society Of Testing And Materials (ASTM) procedures as noted. The performance criteria for use as an anchoring system for threaded rod and reinforcing bars is described in the sections that follow.

52.3.1 Power-Fast™ Physical Property Data

Shelf Life	2 years for components
Storage Conditions	Store dry at 40° to 90° F.
Injection Temperature	Condition material to 60° F minimum
Color	Component A - White Component B - Dark Gray Mixed Epoxy - Light Gray
Mixing Ratio	1 to 1
Consistency	Smooth, non-sag paste
Shore Hardness (ASTM D-2240)	D 86 - 90
Compressive Strength (ASTM D-695)	10,000 psi
Tensile Strength (ASTM-D638)	4,700 psi, Fast Set 5,400 psi, Slow Set
Flexural Strength (ASTM D-790)	6,200 psi, Fast Set 6,700 psi, Slow Set
Slant Shear Strength (ASTM D-732)	4,900 psi, Fast Set - 1 day 6,700 psi, Slow Set - 14 days
Water Absorption (ASTM D-570)	0.59%
Bond Strength (Dry Cure) (ASTM C-882)	3,000 psi, concrete to concrete. 3,000 psi, steel to concrete,
Shrinkage (ASTM D-2566)	Not measurable

52.3.2 Power-Fast Epoxy Injection Gel Setting Time

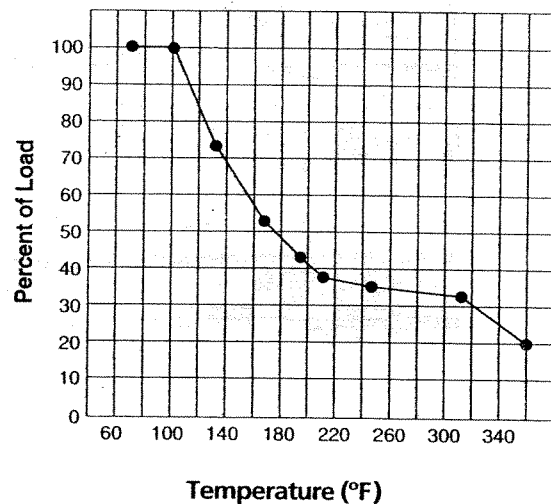
The setting times listed for the **Power-Fast** epoxy vary according to the volume of epoxy used and the base material temperature. The gel time is the maximum time during which the epoxy can be dispensed before it begins to set. At the minimum curing time, the epoxy can support a design load, however, this should be verified by job site testing. The full curing time is the minimum time required for the epoxy to reach its published physical properties.

Base Material Temp.(F°)	Maximum Gel Time (minutes)		Minimum Curing Time (hours)		Full Curing Time (hours)	
	Fast Set	Slow Set	Fast Set	Slow Set	Fast Set	Slow Set
40	30	60	8	16	36	48
60	20	45	3	7	24	36
75	15	35	2	6	24	24
90	10	20	1-1/2	4	16	24

Do not install in base material whose temperature is less than 40°F. unless job site testing is performed.

52.3.3 Effect Of Elevated Temperature

As with all adhesive anchors, the bond strength of the **Power-Fast** epoxy is effected by elevated temperatures in the base material. As the temperature of the base material increases, the bond strength of the anchor will decrease. Typical performance of the **Power-Fast** epoxy at elevated base material temperatures is shown below. The values are based on maintaining the concrete test samples used at a given temperature for a minimum of 24 hours before applying a test load according to ASTM Standard E 1512.



The testing described above is based on heating the entire concrete test element. During a fire, the actual behavior of the **Power-Fast** Epoxy Injection Gel depends upon the heat dissipation inside the concrete. The rate of heat dissipation will vary depending upon the concrete mix design along with the shape and thickness of the structural member. Normally, at depths beyond 3", the concrete heats up relatively slowly since only one face of a slab is actually exposed to the fire.

In addition to the heat dissipation within the base material, the transfer of heat into the adhesive by the anchor rod should be considered. One method of reducing the transfer of heat energy is to protect the fixture and the head of the anchor with a suitable coating. Stainless steel anchor rods typically have a low degree of heat transfer and can be used in conjunction with the **Power-Fast** Epoxy Injection Gel to provide an anchor which has a degree of fire resistance. Testing conducted with Type 316 stainless steel rods installed in normal weight concrete has shown that the **Power-Fast** material can sustain a design load for at least 30 minutes when exposed to a temperature of 1,500° F. Contact Powers Rawl for details of the test program.

52.3.4 Resistance Of Power-Fast Epoxy To Chemicals

The resistance of the cured **Power-Fast** epoxy to various chemicals was determined according to ASTM Specification C-581 by laying molded samples of the resin in the respective chemical agents. The samples were subjected to a bending strength test before and after a 12 month exposure to the chemicals. The epoxy was rated as resistant if there was no visible deterioration and less than 25% reduction in bending strength. This exposure is extreme. Under normal installation conditions, the epoxy is exposed to the chemical agents only at the surface of the concrete around the top of the anchor hole.

Chemical Agent	Concentration	Resistant	Not Resistant
Acetic Acid	40		●
Acetone	10		●
Ammonia	5	●	
Aniline	100		●
Beer	100	●	
Benzine	100	●	
Benzole	100		●
Boric acid, aqueous solution		●	
Calcium carbonate, suspended in water	All	●	
Calcium chloride, suspended in water		●	
Calcium hydroxide, suspended in water		●	
Carbon tetrachloride	100	●	
Caustic soda solution	40		●
Citric acid	All	●	
Chlorine	All	●	
Diesel oil	100	●	
Ethyl alcohol, aqueous solution	50		●
Formaldehyde	30	●	
Formic Acid	100		●
Formic Acid	10	●	
Freon		●	
Fuel oil		●	
Glycol (Ethylene Glycol)		●	
Hydrogen peroxide	30		●
Hydrochloric acid	Conc.	●	
Isopropyl alcohol	100		●
Lactic acid	All	●	
Laitance		●	
Linseed oil	100	●	
Lubricating oil	100	●	
Magnesium chloride, aqueous solution	All	●	
Methanol	100		●

Chemical Agent	Concentration	Resistant	Not Resistant
Motor Oil (SAE 20 W-50)	100	●	
Nitric acid	30		●
Nitric acid	10	●	
Oleic acid	100	●	
Perchloroethylene	100	●	
Petroleum	100	●	
Phenol, aqueous solution	8		●
Phosphoric acid	85	●	
Phosphoric acid	10	●	
Potash Lye (Potassium Hydroxide, 10% and 40% solutions)		●	
Potassium carbonate, aqueous solution	All	●	
Potassium Chlorite, aqueous solution	All	●	
Potassium nitrate, aqueous solution	All	●	
Premium gasoline	100	●	
Sodium carbonate, aqueous solution	All	●	
Sodium chloride, aqueous solution	All	●	
Sodium phosphate, aqueous solution	All	●	
Sodium silicate	All	●	
Sulfuric acid	30		●
Tartaric acid	All	●	
Tetrachloroethylene	100	●	
Toluene			●
Turpentine	100	●	
Trichloroethylene	100		●

52.3.5 Power-Fast Quality Control Procedures

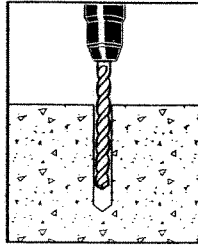
Each **Power-Fast** cartridge is packaged in an individual box on which an identification number is printed to indicate the date of packaging and to provide traceability of the components back to the original manufacturing batch. **Power-Fast** Sausages are individually marked with a similar identification number. Every batch of material is subjected to extensive physical and chemical property testing during manufacture. Each combination of base resin and hardener material batches is tested as an installed anchor to insure that the proper bond strength is developed.

These procedures insure consistent, top quality performance to the specifier and the installer of the **Power-Fast** Epoxy Injection Gel.

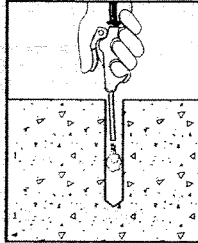
52.8 Power-Fast™ Installation Guidelines

52.8.1 Threaded Rods or Reinforcing Bars in Solid Base Materials

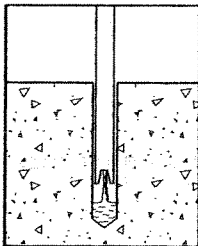
Drill a hole to the size and embedment required. The tolerances of the drill bit used should meet the requirements of ANSI Standard B212.15.



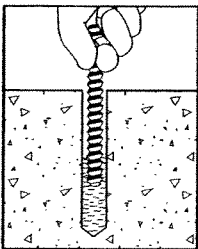
Blow the hole clean with compressed air, brush the hole, and blow it clean again. Holes should be clean and sound. They may be dry or damp, but should be free of standing water or frost. If using reinforcing bars, be sure the bar will fit into the drilled hole. If a larger hole is required, the diameter should be as close as possible to the diameter of the reinforcing bar.



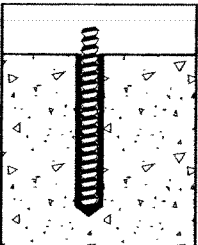
Be sure to properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the hole approximately half way with epoxy starting from the rear of the hole. Slowly withdraw the static mixing nozzle as the hole fills to avoid creating air pockets within the hole.



Push the threaded rod or reinforcing bar into the hole while turning slightly to insure positive dispersion of the epoxy. Be sure the rod is fully seated at the bottom of the hole and that some epoxy has flowed from the top of the hole. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.

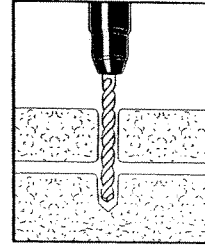


Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.

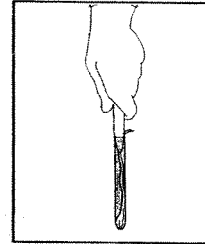


52.8.2 Threaded Rods in Hollow Base Materials

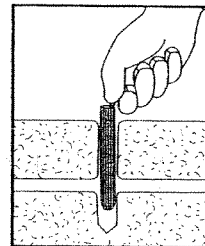
Drill a hole to the size and embedment for the required screen size. The tolerances of the drill bit used should meet the requirements of ANSI B212.15. Blow the hole clean with compressed air, brush the hole, and blow it clean again.



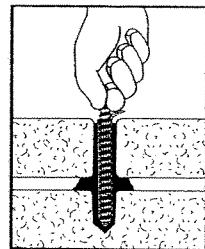
Properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the screen tube with Power-Fast epoxy starting from the rear of the screen. The screen tube should be filled prior to inserting it in the hole.



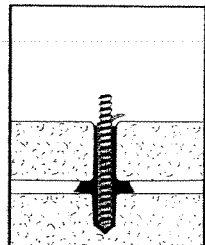
Insert the filled screen tube into the anchor hole until it is fully seated at the required embedment.



Push the threaded rod into the screen while turning slightly to insure positive dispersion of the epoxy. Be sure the rod is fully inserted to the end of the screen tube. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.



Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.



Prior to use, read the box and carton label, Design Manual, Material Data Safety Sheet, and injection tool instructions.

52.9.6 Guide Load Capacities in Masonry Walls

The following tables list the ultimate or failure load capacities for the **Power-Fast** epoxy when tested with high strength anchor rod. The data contained in these tables should be used as a guide since the consistency of masonry base materials varies greatly. Job site tests should be conducted to determine actual load capacities.

52.9.6.1 Ultimate Load Capacities In Red Brick

The following loads are based on tests conducted using threaded rods and screen tubes installed in a multiple wythe constructed from solid red brick. The embedment depth listed indicates the length of screen tube used.

Anchor Size (inches)	Drill Dia. (inches)	Embed. Depth (inches)	Solid Red Brick	
			Tension (lbs.)	Shear (lbs.)
1/4	3/8	2	2,730	1,830
1/4	3/8	3-1/2	3,020	1,830
3/8	1/2	3-1/2	4,395	4,500
3/8	1/2	6	7,430	4,500
3/8	1/2	10	10,660	4,500
1/2	5/8	3-1/2	8,905	6,455
1/2	5/8	6	11,250	6,455
1/2	5/8	10	14,300	6,455
5/8	3/4	3-1/2	8,830	8,470
5/8	3/4	4-1/2	9,435	8,470
5/8	3/4	6	12,580	8,470
5/8	3/4	10	14,600	8,470
3/4	7/8	3-1/2	9,420	10,850
3/4	7/8	6	10,865	10,850
3/4	7/8	8	14,600	10,850
3/4	7/8	13	23,700	10,850

52.9.6.2 Ultimate Load Capacities In C-90 Hollow Block

The following loads are based on tests conducted using threaded rods and screen tubes installed in a wall constructed from C-90 hollow block. The screen tubes were flush with the surface, centered in the block cell.

Anchor Size (inches)	Drill Dia. (inches)	Screen Length (inches)	C-90 Hollow Block	
			Tension (lbs.)	Shear (lbs.)
1/4	3/8	2	1,080	1,740
3/8	1/2	3-1/2	1,830	3,480
1/2	5/8	3-1/2	2,630	4,460
5/8	3/4	4-1/2	3,020	5,320
3/4	7/8	6	4,060	6,040

52.9.6.3 Ultimate Load Capacities In Grout Filled Block

The following loads are based on tests conducted using threaded rods installed in a wall constructed from C 90 hollow block filled with fine grout.

Anchor Size (inches)	Drill Dia. (inches)	Embed. Depth (inches)	Grout Filled Block	
			Tension (lbs.)	Shear (lbs.)
1/4	5/16	1	1,815	1,880
1/4	5/16	2	2,730	1,880
3/8	7/16	1-1/2	4,510	4,500
3/8	7/16	3-1/2	6,910	4,500
1/2	9/16	2	5,160	6,455
1/2	9/16	4-1/4	9,630	6,455
5/8	3/4	2-1/2	8,245	9,460
5/8	3/4	5	13,910	9,460
3/4	7/8	3	8,560	11,840
3/4	7/8	6-5/8	15,940	11,840

52.10 Use in Special Applications

52.10.1 Installation Of Anchors Under Water

Power-Fast epoxy injection gel can be used for the installation of threaded anchor rod or reinforcing bars in submerged applications under water provided some installation and design criteria are followed. The anchor holes should be prepared following the standard installation instructions with the following exceptions.

Special care should be taken to clean the anchor hole as a slurry of concrete paste tends to form on the walls of the anchor hole when drilling under water. To inject the epoxy, insert the mixing nozzle to the bottom or rear of the anchor hole. Slowly withdraw the nozzle as the anchor hole is filled to insure that the water is displaced from the hole and that no air pockets are formed. The anchor hole should be filled completely with epoxy prior to inserting the anchor rod.

The setting time of the **Power-Fast** epoxy in submerged applications depends upon the base material temperature as listed in the material properties section.

Laboratory tests and field experience have shown that a decrease of 15 to 20% in the ultimate tension load capacity can be expected for an anchor which is installed under water. The design professional should include this reduction in his calculations. If the tension capacities are critical, a job site test is recommended.

52.10.2 Installation Of Anchors Overhead

Prior to installing adhesive anchors overhead or in walls, consult with the governing building authority to determine the requirements for fire safety guidelines. The effect of elevated temperature on the **Power-Fast** epoxy is shown in the material properties section.

Power-Fast epoxy injection gel can be used for the installation of threaded anchor rod overhead. The anchor holes should be prepared following the standard installation instructions with the following exceptions.

After the anchor hole is drilled and cleaned, insert the proper size anchor seal plug into the hole until the rim of the plug is fully seated against the base material. Insert the **Power-Fast** mixing nozzle through the cross cut opening of the anchor seal plug. Hold the plug in place and begin filling the anchor hole half way with **Power-Fast** epoxy starting from the bottom or rear of the anchor hole. Insert the threaded rod through the cross cut opening of the anchor seal plug. Turn the rod slightly to insure proper distribution of the epoxy. Be sure that the anchor rod is fully seated at the bottom of the anchor hole and that some epoxy has flowed from the cross cut opening in the anchor seal plug.